

Presentation I: Basics

Physics-Aware Action-Conditioned World Modeling for Billiards

Your Name

March 3, 2026

Why This Project?

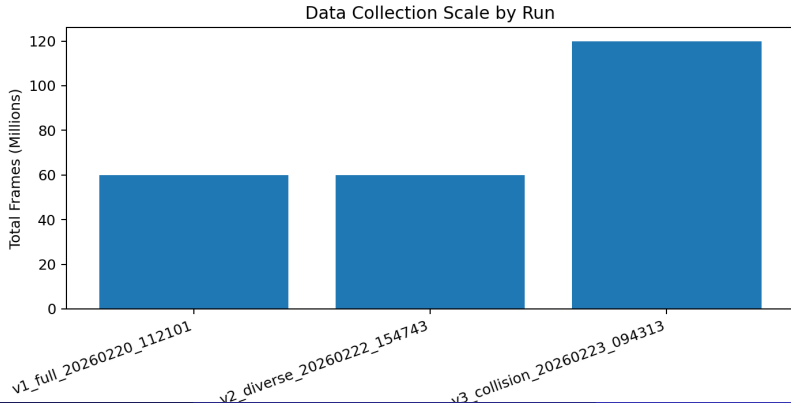
- Goal: generate interactive billiards rollouts conditioned on user actions.
- Key challenge: not just visual quality, but physical logic over time.
- Physics checks:
 - do balls bounce correctly off walls?
 - do collisions look physically plausible?
 - do object identities stay consistent over rollout?

Pipeline Overview

- 1 Data generation from simulator:
 - frames x_t , actions a_t , simulator state s_t
- 2 VAE compresses frames to latents z_t
- 3 Action-conditioned DiT predicts next-latent diffusion noise
- 4 Rollout eval at horizons $h = \{1, 4, 8, 16, 32\}$
- 5 Interactive inference route for click/drag actions

Data Scale ($v1 + v2 + v3$)

- v1: 100,000 episodes (baseline coverage)
- v2: 100,000 episodes (higher diversity)
- v3: 200,000 episodes (collision-heavy)
- 600 frames per episode $\Rightarrow$ 400k episodes total



Diffusion Model Basics

Forward corruption in latent space:

$$x_\tau = \sqrt{\bar{\alpha}_\tau} x_0 + \sqrt{1 - \bar{\alpha}_\tau} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

Model predicts noise $\hat{\epsilon}_\theta$:

$$\mathcal{L}_{\text{base}} = \mathbb{E} \|\hat{\epsilon}_\theta - \epsilon\|_2^2$$

Reconstruct predicted clean latent:

$$\hat{x}_0 = \frac{x_\tau - \sqrt{1 - \bar{\alpha}_\tau} \hat{\epsilon}_\theta}{\sqrt{\bar{\alpha}_\tau}}$$

- Context window: $z_{t-L+1:t}$, with $L = 8$
- Action vector: $a_t = [f_x, f_y, \text{trigger}]$
- Model predicts next-latent diffusion noise conditioned on context + action

$$\hat{\epsilon}_t = f_{\theta}(z_{t-L+1:t}, a_t, x_{\tau}, \tau)$$

- Inference uses DDIM sampling for speed/quality tradeoff.

Diffusion Forcing (Core Idea)

- Baseline trains mostly one-step.
- Rollout inference is multi-step autoregressive.
- Diffusion forcing trains multi-step with scheduled teacher forcing.

$$\mathcal{L}_{\text{DF}} = \frac{1}{K} \sum_{k=1}^K \|\hat{\epsilon}_k - \epsilon_k\|_2^2$$

$$\tilde{z}_{k+1} = m_k z_{k+1}^{gt} + (1 - m_k) \hat{x}_{0,k}, \quad m_k \sim \text{Bernoulli}(p_{\text{tf}}(n))$$

- This reduces train/inference mismatch and helps long-horizon stability.

Evaluation Protocol

- Fixed-horizon rollout metrics at $h = \{1, 4, 8, 16, 32\}$
- Latent metrics: MSE, MAE
- Frame metrics (after VAE decode): PSNR, L1
- Important: metric “@16” means exactly step-16 prediction, not average over 1..16

Takeaway for Deck II

- Deck I covered: data, VAE, diffusion, DiT, DF, and evaluation setup.
- Deck II will answer:
 - Did diffusion forcing actually improve interactive physics?
 - Where does the model still fail?
 - What are the concrete next steps?